# Étienne Fodor

Physics of Active Matter
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## Scientific positions and education

Since 2025 **Associate Professor**, DPhyMS, Univ of Luxembourg

2020–25 **Assistant Professor**, DPhyMS + **ATTRACT Fellow**, Fonds National de la Recherche (FNR)

2017–20 **Oppenheimer Fellow**, Univ of Cambridge + **Junior Research Fellow**, St Catharine's College

2016–17 **Postdoctoral Research Associate**, DAMTP, Univ of Cambridge (Supervisor: ME Cates)

2013–16 **PhD in Physics**, Univ Paris Diderot (Supervisors: P Visco, F van Wijland), summa cum laude
PhD thesis: "Tracking nonequilibrium in living matter and self-propelled systems"

2012–13 **Masters in Physics**, École Normale Supérieure (ENS), Paris

2011–12 **Agrégation de Physique**, Training program for teaching in Physics Bachelors, ENS Cachan

2009–11 **Bachelors + Masters in Physics**, ENS Lyon

## Research and teaching

Since 2022 **Masters course in Physics**, DPhyMS, Univ of Luxembourg | 14 weeks/year
Course: "Nonequilibrium soft and active matter"

2024 **Masters course in Physics**, Dept of Physics, Univ of Liège | 10 hours, 1 week
Course: "Statistical physics of passive and active matter"

2021 **Doctoral course in Physics**, DPhyMS, Univ of Luxembourg | 6 hours, 1 day
Course: "Field theory for soft matter"

2015–16 **PhD exchange program**, YITP, Kyoto Univ (Host: H Hayakawa) | 2 months/year

2013–16 **Bachelors Physics tutorials**, Univ Paris Diderot | 14 weeks/year

2013 **Masters internship**, Univ Paris Diderot (Supervisors: P Visco, F van Wijland) | 16 weeks
Masters thesis: "Active forces in living cells"

2012–13 **Bachelors Physics tutorials**, Lycée Fénelon, Paris | 30 weeks

2011 **Masters intership**, Univ of Oxford (Supervisors: AS Wyatt, IA Walmsley) | 12 weeks
Masters thesis: "Characterization and control of extreme pulses from high harmonic generation"

2010–11 **Bachelors Physics tutorials**, Lycée la Martinière Monplaisir, Lyon | 30 weeks

2010 **Bachelors internship**, Univ de Genève (Supervisors: L Bonacina, J-P Wolf) | 8 weeks
Bachelors thesis: "Micro-spectroscopie CARS à l'aide d'une source laser unique"

## Funding and selected awards

2027 **Outstanding Junior Fellowship**, Institute of Advanced Studies, Tel Aviv Univ

2026 **Emerging Investigator**, Soft Matter + **Trusted Reviewer**, Institute of Physics
**Young Leader**, Science and Technology in Society forum, Kyoto

2026–29 **CORE grant**, **OptimalControlTransition**, FNR, Luxembourg | 800 kEUR
**Doctoral Training Unit**, **BECR**, FNR (Program lead: A Skupin)
**Doctoral Training Network**, **CAFE-BIO**, Horizon Europe (Program lead: R Blythe)

2025 **Lorentz Center Workshop**, **Nonequilibrium Systems Under Control**, Leiden | 30 kEUR

2024–27 **CORE grant**, **ActiveTopo**, FNR, Luxembourg | 830 kEUR

2021–26 **Doctoral Training Unit**, **ACTIVE**, FNR (Program lead: M Esposito)

2020–25 **ATTRACT Fellowship**, FNR | 1.5 MEUR

2017–20 **Oppenheimer + Junior Research Fellowship**, St Catharine's College, Univ of Cambridge

2017 **PhD prize**, Institut Systèmes Complexes, Paris + **Best talk**, SIAM-IMA Conference, Cambridge

2015 **Best talk**, Active Liquids, Lorentz Center, Leiden

2013–16 **Teaching Assistantship**, Univ Paris Diderot + **PhD Scholarship**, ENS Cachan

## Supervision and mentoring

Since 2020 **Principal Investigator**, Head of group "Physics of active matter", Univ of Luxembourg

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|---|---|---|
| 10 Postdocs | Since 2026 | JC Sunil + M Scandolo |
| | Since 2025 | A Soriani + F Thewes |
| | 2024–25 | F Serafin (after: Postdoc, Univ of Luxembourg) |
| | 2023–25 | UA Dattani (after: Deutsche Bank, Berlin) |
| | 2022–26 | T Banerjee, CORE Junior Fellow (after: Faculty member, NISER, India) |
| | 2022–25 | WD Piñeros, MSCA Fellow (after: Postdoc, CEA Caen, France) |
| | 2021–24 | A Manacorda, MSCA Fellow (after: Postdoc, Sapienza Univ, Roma) |
| | 2020–22 | LK Davis (after: Flora Philip Research Fellow, Univ of Edinburgh) |
| 8 PhDs | Since 2026 | A Jagannathan + N Di Serio |
| | Since 2024 | N Setzkorn + IJC Miranda + M Antonioli |
| | Since 2023 | L Casagrande, AFR Fellow (Luxembourg) |
| | 2021–25 | BN Radhakrishnan (after: Postdoc, Western Univ, Canada) |
| | 2020–23 | Y Zhang (after: United Consulting Group, Frankfurt) |
| 4 Masters | 2025–26 | A Jagannathan |
| | 2023–24 | N Setzkorn |
| | 2022–23 | L Casagrande + T Desaleux |

2016–20 **Student co-supervisor**, DAMTP, Univ of Cambridge (supervisor: ME Cates)

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|---|---|
| 2 PhDs | ØL Borthne (2017–20) + T Ekeh (2018–21) |
| 3 Masters | D Martin (2016–17) + T Ekeh (2017–18) + JW Knight (2019–20, best thesis prize) |

## PhD committees

2026 **Univ of Luxembourg**, A Kabylda* (supervisor: A Tkatchenko)
**Univ of Barcelona**, Y Rouzaire (supervisor: D Levis)
**Tata Institute of Fundamental Research**, A Bansal (supervisor: M Rao)

2025 **Univ of Luxembourg**, SGM Srinivas* (supervisor: M Esposito)
**Univ of Mons**, G Palumbo (supervisor: P Damman)
**Univ of Luxembourg**, M Puleva* (supervisor: A Tkatchenko)

2024 **Univ Grenoble Alpes**, L Guislain (supervisor: E Bertin)
**Univ Paris Cité**, A Dinelli, (supervisor: J Tailleur)
**Univ of Luxembourg**, N Carabba* + L Dupays (supervisor: A del Campo)

2023 **Imperial College London**, Z Zhang (supervisor: G Pruessner)

2022 **Univ of Luxembourg**, E Penocchio + D Forastiere (supervisor: M Esposito)
**Univ of Luxembourg**, V Vassilev Galindo (supervisor: A Tkatchenko)

2021 **Luxembourg Centre for Systems Biomedicine**, S Martina (supervisor: A Skupin)
**Univ of Luxembourg**, J Ekström* (supervisor: TL Schmidt)

* Head of jury for these PhD defences

## Review and outreach activities

Since 2025 **Editorial board member**, Journal of Statistical Mechanics

Since 2016 **Reviewer for journals and agencies** | ca 30 reviews/year
Selected journals | Commun Phys, J Chem Phys, J Phys A, J Stat Mech, Nat Commun, Nat Phys, New J Phys, Newton, Phys Rev (E, Lett, Res, X), PNAS, Science, Science Adv
Selected agencies | ANR (France), CECAM (Switzerland), DFG (Germany), DOE (USA), ERC + MSCA (Horizon Europe), FNRS-FRS (Belgium), ISF (Israel)

2026 **Guest editor**, Physical Review E | Special topics: "Controlling stochastic dynamics across scales"
Co-editor: TR Gingrich (Northwestern Univ)

2026 **Guest editor**, New Journal of Physics | Special topics: "Statistical mechanics of active matter"
Co-editors: M Fruchart (ESPCI), SAM Loos (MPI Goettingen), T Markovich (Tel Aviv Univ)

2024 **Outreach lecture**, Institut Etudes Scientifiques, Cargèse

2023 **Inaugural lecture**, Faculty of Science, Technology and Medicine, Univ of Luxembourg

2021–24 **High-school seminars and intern hosts + Student fairs**, DPhyMS, Univ of Luxembourg

## Organized events

2026 **Nonequilibrium Statistical Mechanics**, Univ Paris Cité, Paris | Workshop, 1 day
Co-organizer: F van Wijland (Univ Paris Cité)

2025 **Nonequilibrium Systems Under Control**, Lorentz Center, Leiden | Workshop, 1 week
Co-organizers: TR Gingrich (Northwestern Univ), SAM Loos (Univ of Cambridge)

2024 **Energy, Information and Evolution in Biology**, Cargèse | Summer school, 2 weeks
Co-organizers: A Manacorda, M Esposito (Univ of Luxembourg)
**Physics Meets Mathematics**, Univ of Luxembourg | Workshop, 1 day

2018–20 **Statistical Physics and Soft Matter**, DAMTP, Univ of Cambridge | Weekly seminar
Co-organizers: ME Cates, RL Jack (Univ of Cambridge)

2019 **Colloids as a Toolbox for Statistical Mechanics**, Univ of Cambridge | Workshop, 1 day
Co-organizers: ME Cates, RL Jack (Univ of Cambridge)

2018 **Nonequilibrium Biophysics**, **World Congress of Biomechanics**, Dublin | Session, $\frac{1}{2}$ day
Co-organizer: D Mizuno (Kyushu Univ)

## Presentation in conferences and seminars

Recording of selected presentations: efodorphysics.github.io/talks.html

### Invited presentations in conferences

2027 **Mechanics and Synchronization in Dense Active Matter**, Cargèse

2026 **Nonequilibrium Dynamics, from Active Matter to Evolutionary Dynamics**, KITP
**Control and Optimisation of Out-of-equilibrium Processes**, Beg Rohu
**SigmaPhi2026**, **Statistical Physics of Complex Systems**, Kolymbari
**Frontiers in Statistical Physics and Soft Matter**, Edwards Symposium, Univ of Cambridge

2025 **Statistical Physics of Living Systems**, CECAM, Lausanne
**International Discussion Meeting on Relaxations in Complex Systems**, Barcelona
**Self-Organization Far From Equilibrium**, APS March meeting, Anaheim
**Machine Learning for Enhanced Sampling of Atomistic Systems**, Berkeley

2024 **The Many Faces of Active Mechanics**, KITP, Santa Barbara
**Nonequilibrium Statistical Physics of Complex Systems**, Seoul

2023 **Computational Advances in Active Matter**, Lorentz Center, Leiden
**Frontiers in Nonequilibrium Physics: Active Matter, Topology and Beyond**, Kyoto
**Conference on Statistical Mechanics**, Sitges
**Physics of Dense and Active Disordered Materials**, Kyoto
**Frontiers in Nonequilibrium Physics**, Institute of Mathematical Sciences, Chennai

2022 **Statistical Mechanical Theories of Emergence in Biological Systems**, Edinburgh
**Numerical Techniques for Nonequilibrium Steady States**, CECAM, Mainz

2020 **Symmetry, Thermodynamics and Topology in Active Matter**, KITP (online)

2018 **Why Measure Entropy Production?**, Princeton Univ
**Active Matter Session**, Berkeley

### Contributed presentations in conferences

2026 **DPG Spring Meeting**, **Active Matter V**, Dresden

2025 **Multiscale Modeling of Chemically Active Mixtures**, CECAM, Lausanne
**StatPhys29**, **Out-of-equilibrium Statistical Physics**, Florence
**From Flocking Birds to Migrating Cells: Recent Advances in Active Matter**, Leiden

2024 **Dissipative Processes in Molecular Systems**, Padova
**Workshop on Stochastic Thermodynamics V** (online)
**DPG Spring Meeting**, **Stochastic Thermodynamics**, Berlin

2023 **StatPhys28**, **Soft Matter**, Tokyo
**Bridge between Non-equilibrium Statistical Physics and Biology**, Cambridge
**New Perspectives in Active Systems**, Dresden
**From Soft Matter to Biophysics**, Les Houches

2021 **Liquid Matter Conference**, Prague (online)
**Workshop on Stochastic Thermodynamics II** (online)

2020 **Motile Active Matter Conference**, Bonn (online)

2019 **StatPhys27**, **Out-of-equilibrium aspects**, Buenos Aires
**International Soft Matter Conference**, Edinburgh
**Statistical Physics of Complex Systems**, Nordita, Stockholm

2018 **Nonequilibrium Collective Dynamics**, Technische Univ Berlin
**Fundamental Problems in Active Matter**, Aspen Center for Physics

2017 **SIAM-IMA Annual Conference**, Univ of Cambridge
**Edwards Centre Mini Conference**, Univ of Cambridge
**Open Statistical Physics**, Milton Keynes

2016 **StatPhys26**, **Biological Physics**, Lyon
**Non-Gaussian Workshop**, YITP, Kyoto

2015 **Active Liquids**, Lorentz Center, Leiden

2014 **Condensed Matter in Paris**, Univ Paris Descartes
**Journées de Physique Statistique**, Paris

**Invited seminars and colloquia**

2027 **Physics Dept Seminar**, Technion
**Institute of Advanced Studies Colloquium**, Tel Aviv University

2026 **Center for Data Science and Complexity Colloquium**, Univ of Münster
**Institut für Theoretische Physik Seminar**, Univ of Münster
**Gulliver Lab Seminar**, ESPCI, Paris
**MPI Physics of Complex Systems Seminar**, Dresden
**Dept of Physics Seminar**, Univ of Oxford
**YITP Seminar**, Kyoto
**Theoretical Biophysics Seminar**, Curie Institute, Paris

2025 **Soft Condensed Matter Seminar**, Harvard University
**Biophysics Seminar**, Massachusetts Institute of Technology
**Solitons at Work Seminar** (online)

2024 **Dept of Chemistry Seminar**, Univ of California, Berkeley
**LPTMC Seminar**, Sorbonne Univ, Paris
**Dept of Physics Seminar**, Univ of Liège
**Institute of Physics Seminar**, Univ of Leiden
**Niels Bohr Institute Seminar**, Univ of Copenhagen

2023 **Biological, Soft and Complex Materials and Theory Seminar**, Univ of Bristol
**EMBL Theory Seminar**, Heidelberg

2022 **Biological Physics and Physical Biology Seminar** (online)
**Soft Matter Seminar**, DAMTP, Univ of Cambridge (online)
**Mathematical Physics Seminar**, Imperial College London (online)

2021 **Dept of Physics**, Guangdong Technion (online)
**Quantum Science and Technology Seminar**, Univ of Luxembourg (online)
**Non-equilibrium Statistical Physics Seminar**, Georg-August-Univ Göttingen (online)
**Centre de Physique Théorique Seminar**, Aix-Marseille Univ (online)

2020 **School of Physics and Astronomy Seminar**, Univ of Edinburgh (online)
**Dept of Physics Seminar**, Univ of Bath

2019 **Quantitative Life Sciences Seminar**, ICTP, Trieste
**Dept of Chemistry Seminar**, James Franck Institute, Univ of Chicago
**Physics of Living Systems Seminar**, Massachusetts Institute of Technology
**DPhyMS Seminar**, Univ of Luxembourg
**Institute of Physics Seminar**, Univ of Amsterdam

2018 **LiPhy Seminar**, Univ Grenoble Alpes

**Charles Coulomb Lab Seminar**, Univ de Montpellier
**Gulliver Lab Seminar**, ESPCI, Paris
**Graduate Research Seminar**, St Catharine's College, Cambridge
**Research Colloquium**, California State Univ, Fullerton

2017 **Soft Matter Seminar**, DAMTP, Univ of Cambridge
**BioLunch Seminar**, DAMTP, Univ of Cambridge

2016 **School of Mathematical Sciences Seminar**, Queen Mary Univ of London
**Soft Matter Seminar**, DAMTP, Univ of Cambridge
**MSC Lab Seminar**, Univ Paris Diderot
**YITP Seminar**, Kyoto

2015 **LiPhy Seminar**, Univ Grenoble Alpes
**Physics-Biology Interface Seminar**, Univ Paris Sud
**Soft Matter Seminar**, DAMTP, Univ of Cambridge
**YITP Seminar**, Kyoto

2014 **MSC Lab Seminar**, Univ Paris Diderot

## Scientific manuscripts

[52] **Collective order reorganizes dissipation and powers an active engine**
M Riedl, V Wattiez, ÉF, J Brugués, and F Romanò, arXiv:2609.31059

[51] **Active pistons extract work by periodic compression alone**
P Bernard, E Kwon, B Buhl, ÉF, and O Dauchot, arXiv:2609.15380

[50] **Topology of pulsating active matter: Defect asymmetry controls emergent motility**
L Casagrande, A Manacorda, and ÉF, arXiv:2605.25996

[49] **Defect statistics in pulsating active liquids: From contraction waves to aster dynamics**
T Banerjee, L Cocconi, T Desaleux, J Ranft, and ÉF, arXiv:2509.19024

[48] **Control of active field theories at minimal dissipation**
A Soriani, E Tjhung, ÉF, and T Markovich, arXiv:2504.19285

[47] **Hydrodynamics of pulsating active liquids**
T Banerjee, T Desaleux, J Ranft, and ÉF, arXiv:2407.19955

[46] **Collective deformation of anisotropic particles with internal pulsation**
L Casagrande, A Manacorda, and ÉF, J Phys A: Math Theor **59**, 405001 (2026)

[45] **Irreversibility in scalar active turbulence: The role of topological defects**
BN Radhakrishnan,* F Serafin,* TL Schmidt, and ÉF, New J Phys **28**, 034601 (2026)

[44] **Entropy production rate in thermodynamically consistent flocks**
T Agranov, RL Jack, ME Cates, and ÉF, New J Phys **27**, 104602 (2025)

[43] **Quantifying dissipation in flocking dynamics: When tracking internal states matters**
K Proesmans, G Falasco, A Tanaji Mohite, M Esposito, and ÉF, Phys Rev E **112**, 024103 (2025)

[42] **Diffusive oscillators capture the pulsating states of deformable particles**
A Manacorda and ÉF, Phys Rev E **111**, L053401 (2025)

[41] **Species interconversion of deformable particles yields transient phase separation**
Y Zhang, A Manacorda, and ÉF, New J Phys **27**, 043023 (2025)

[40] **Biased ensembles of pulsating active matter**
WD Piñeros and ÉF, Phys Rev Lett **134**, 038301 (2025) | Editors' suggestion

[39] **Nonequilibrium thermodynamics of non-ideal reaction-diffusion systems: Implications for active self-organization**
F Avanzini, T Aslyamov, ÉF, and M Esposito, J Chem Phys **161**, 174108 (2024)

[38] **Controlling active matter: The need for thermodynamic consistency**
ÉF, Europhys News **55**, 20 (2024)

[37] **Thermodynamically consistent flocking: From discontinuous to continuous transitions**
T Agranov, RL Jack, ME Cates, and ÉF, New J Phys **26**, 063006 (2024)

[36] **Active matter under control: Insights from response theory**
LK Davis, K Proesmans, and ÉF, Phys Rev X **14**, 011012 (2024) | Highlight in Physics 17, 20 (2024)

[35] **Pulsating active matter**
Y Zhang and ÉF, Phys Rev Lett **131**, 238302 (2023)

[34] **Non-ideal reaction-diffusion systems: Multiple routes to instability**
T Aslyamov, F Avanzini, ÉF, and M Esposito, Phys Rev Lett **131**, 138301 (2023)

[33] **Towards a liquid-state theory for active matter**
YI Li, R Garcia-Millan, ME Cates, and ÉF, EPL **142**, 57004 (2023)

[32] **Thermodynamic control of activity patterns in cytoskeletal networks**
A Lamtyugina, Y Qiu, ÉF, AR Dinner, and S Vaikuntanathan, Phys Rev Lett **129**, 128002 (2022)

[31] **From predicting to learning dissipation from pair correlations of active liquids**
G Rassolov, L Tociu, ÉF, and S Vaikuntanathan, J Chem Phys **157**, 054901 (2022)

[30] **Mean-field theory for the structure of strongly interacting active liquids**
L Tociu, G Rassolov, ÉF, and S Vaikuntanathan, J Chem Phys **157**, 014902 (2022)

[29] **Power fluctuations in sheared amorphous materials: A minimal model**
T Ekeh, ÉF, SM Fielding, and ME Cates, Phys Rev E **105**, L052601 (2022)

[28] **Irreversibility and biased ensembles in active matter: Insights from stochastic thermodynamics**
ÉF, RL Jack, and ME Cates, Annu Rev Condens Matter Phys **13**, 215 (2022)

[27] **Stochastic hydrodynamics of complex fluids: Discretisation and entropy production**
ME Cates, ÉF, C Nardini, T Markovich, and E Tjhung, Entropy **24**, 254 (2022) | Editor's choice

[26] **Optimal power and efficiency of odd engines**
ÉF and A Souslov, Phys Rev E **104**, L062602 (2021)

[25] **Thermodynamics of active field theories: Energetic cost of coupling to reservoirs**
T Markovich, ÉF, E Tjhung, and ME Cates, Phys Rev X **11**, 021057 (2021)

[24] **Active engines: Thermodynamics moves forward**
ÉF and ME Cates, EPL **134**, 10003 (2021)

[23] **Statistical mechanics of active Ornstein-Uhlenbeck particles**
D Martin, J O'Byrne, ME Cates, ÉF, C Nardini, J Tailleur, and F van Wijland,
Phys Rev E **103**, 032607 (2021)

[22] **Collective motion in large deviations of active particles**
Y-E Keta, ÉF, F van Wijland, ME Cates, and RL Jack, Phys Rev E **103**, 022603 (2021)

[21] **Time-reversal symmetry violations and entropy production in field theories of polar active matter**
ØL Borthne, ÉF, and ME Cates, New J Phys **22**, 123012 (2020)

[20] **Thermodynamic cycles with active matter**
T Ekeh, ME Cates, and ÉF, Phys Rev E **102**, 010101(R) (2020)

[19] **Dissipation controls transport and phase transitions in active fluids: Mobility, diffusion and biased ensembles**
ÉF, T Nemoto, and S Vaikuntanathan, New J Phys **22**, 013052 (2020)

[18] **Autonomous engines driven by active matter: Energetics and design principles**
P Pietzonka, ÉF, C Lohrmann, ME Cates, and U Seifert, Phys Rev X **9**, 041032 (2019)

[17] **How dissipation constrains fluctuations in nonequilibrium liquids: Diffusion, structure and biased interactions**
L Tociu, ÉF, T Nemoto, and S Vaikuntanathan, Phys Rev X **9**, 041026 (2019)

[16] **Driven probe under harmonic confinement in a colloidal bath**
V Démery and ÉF, J Stat Mech **2019**, 033202 (2019)

[15] **Optimizing active work: Dynamical phase transitions, collective motion and jamming**
T Nemoto, ÉF, ME Cates, RL Jack, and J Tailleur, Phys Rev E **99**, 022605 (2019)

[14] **Non-Gaussian noise without memory in active matter**
ÉF, H Hayakawa, J Tailleur, and F van Wijland, Phys Rev E **98**, 062610 (2018)

[13] **The statistical physics of active matter: From self-catalytic colloids to living cells**
ÉF and MC Marchetti, Physica A **504**, 106 (2018)

[12] **Extracting maximum power from active colloidal heat engines**
D Martin, C Nardini, ME Cates, and ÉF, EPL **121**, 60005 (2018) | Editor's choice

[11] **Active mechanics reveal molecular-scale force kinetics in living oocytes**
WW Ahmed,* ÉF,* M Almonacid,* M Bussonnier, NS Gov, M-H Verlhac, P Visco, F van Wijland, and T Betz, Biophys J **114**, 1667 (2018)

[10] **Spatial fluctuations at vertices of epithelial layers: Quantification of regulation by Rho pathway**
ÉF,* V Mehandia,* J Comelles, R Thiagarajan, NS Gov, P Visco, F van Wijland, D Riveline
Biophys J **114**, 939 (2018)

[9] **Entropy production in field theories without time-reversal symmetry: Quantifying the non-equilibrium character of active matter**
C Nardini, ÉF, E Tjhung, F van Wijland, J Tailleur, and ME Cates, Phys Rev X **7**, 021007 (2017)

[8] **Nonequilibrium dissipation in living oocytes**
ÉF,* WW Ahmed,* M Almonacid,* M Bussonnier, NS Gov, M-H Verlhac, T Betz, P Visco,
and F van Wijland, EPL **116**, 30008 (2016)

[7] **How far from equilibrium is active matter?**
ÉF, C Nardini, ME Cates, J Tailleur, P Visco, and F van Wijland, Phys Rev Lett **117**, 038103 (2016)
Editor's suggestion | Highlight in Physics 9, s76 (2016)

[6] **Active cage model of glassy dynamics**
ÉF, H Hayakawa, P Visco, and F van Wijland, Phys Rev E **94**, 012610 (2016)

[5] **Modeling the dynamics of a tracer particle in an elastic active gel**
E Ben Isaac, ÉF, P Visco, F van Wijland, and NS Gov, Phys Rev E **92**, 012716 (2015)

[4] **Active cell mechanics: Measurement and theory**,
WW Ahmed, ÉF, and T Betz, Biochimica et Biophysica Acta - Mol Cell Res **1853**, 3083 (2015)

[3] **Activity-driven fluctuations in living cells**
ÉF,* M Guo,* NS Gov, P Visco, DA Weitz, and F van Wijland, EPL **110**, 48005 (2015)
Editor's choice | Highlight in Europhysics News 46/5 (2015)

[2] **Generalized Langevin equation with hydrodynamic backflow: Equilibrium properties**
ÉF, DS Grebenkov, P Visco, and F van Wijland, Physica A **422**, 107 (2015)

[1] **Energetics of active fluctuations in living cells**
ÉF, K Kanazawa, H Hayakawa, P Visco, and F van Wijland, Phys Rev E **90**, 042724 (2014)

* Equal contribution of these authors to this work